Quick Work
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Tutorial-12. (Boost Strap)

Q.

Sample	feature 1 (Age)	feature 2 (Income)	feature 3 (Credit Score)	Target (Loan Approved)
S1	25	40	650	Yes
S2	30	60	700	Yes
S3	45	35	600	No
S4	50	30	580	No
S5	28	55	720	Yes

Sol:-

Boost Strap → Sampling with Replacement.

1> Boostrap 1 (B1)

S1, S2, S3, S4, S5 → Replacing S3 with S2.

Boostrap 2 (B2)

S1, S2, S3, S4, S5 → S2, S3, S4, S4, S5

Boostrap 3 (B3)

S1, S2, S3, S4, S5 → S1, S2, S3, S3, S4

2> Decision Tree (B1)

S1, S2, S3, S4, S5
↓ ↓ ↓ ↓ ↓
Yes Yes Yes No Yes

Age
≤30 → Yes
>30 → No

} → This is called single split decision Tree.

Decision Tree (B2)

S2, S3, S4, S4, S5
↓ ↓ ↓ ↓ ↓
Yes No No No Yes

Credit Score

≥ 650 → Yes

< 650 → No

Decision Tree (B3)

S1, S2, S3, S3, S4

↓ ↓ ↓ ↓ ↓

Yes Yes No No No

Income

≥ 40 → Yes

< 40 → No.

3&4) Prediction & Voting

Age = 32, Income = 50, Credit Score = 690.

↓ ↓ ↓

Loan approval :- No Yes Yes.

Majority Voting :- No, Yes, Yes.

↓

Y.

ii) Age = 35, Income = 45, Credit Score = 750

↓ ↓ ↓

No Yes Yes.

Majority Voting :- No, Yes, Yes

↓

Yes